

OEM MAINTENANCE AND OPERATIONS MANUAL

Multi-Line Production Equipment Knowledge Base for Telemetry- and Schedule-Aware Predictive Maintenance

Document type	Controlled reference
OEM Maintenance and Operations Manual	AI/RAG knowledge base for factory telemetry and IT batch schedules
Lines covered	LINE-1-MIXING; LINE-2-PACKAGING; LINE-3-FILLING; LINE-4-SEALING; LINE-5-PALLETIZING
Primary use	Maintenance diagnosis, operations context, predictive-maintenance recommendations, and schedule-aware retrieval
Revision	V1.0 - consolidated multi-line knowledge-base edition

Important: This manual combines the user-defined cross-line telemetry guardrails with the detailed equipment context present in the uploaded LINE-2-PACKAGING OEM source. Where the uploaded source contains line-specific values that differ from the required cross-line guardrails, both are preserved and explicitly labeled so an AI retrieval system can distinguish the contexts rather than overwrite one with the other.

Prepared as an OEM-style technical reference. Site-specific safety procedures, controlled drawings, torque records, lubrication specifications, and revision-controlled PLC/HMI logic remain the authority for physical maintenance execution.

Document Control and Use

This manual is organized for both human maintenance personnel and retrieval-augmented AI agents. Each production line contains the same four required operational sections followed by maintenance, diagnostic, and metadata guidance. Exact line IDs are preserved throughout the document for deterministic retrieval.

Scope

The knowledge base covers five production lines: LINE-1-MIXING, LINE-2-PACKAGING, LINE-3-FILLING, LINE-4-SEALING, and LINE-5-PALLETIZING. The primary monitoring signals are temperature and vibration, with batch SKU, machine changeover state, throughput, and Predicted RUL used as contextual features.

Safety and execution boundary

Predictive recommendations are decision-support outputs. Physical intervention requires the site-approved risk assessment, isolation procedure, lockout/tagout (LOTO), competent-person authorization, and applicable electrical, pneumatic, hydraulic, thermal, and mechanical safety controls. An AI recommendation must not bypass an interlock, defeat a guard, or instruct a worker to touch energized or moving equipment.

Telemetry interpretation principle

Telemetry must be interpreted with IT batch schedule context. The same vibration or temperature signal can mean different things during startup, steady-state production, SKU-899 heavy-duty processing, and the typical 30-minute Machine Changeover window. The RAG agent should retrieve both the active line section and the active batch context before recommending action.

Source context used in this edition

The uploaded reference is a two-page document titled **OEM MAINTENANCE & OPERATING MANUAL**, document ID **OEM-MAN-L2P-2025-V2**, for **LINE-2-PACKAGING**. Its first page contains SKU-specific packaging operating parameters plus diagnostic errors E-104 and E-208; its second page contains E-312, weekly/monthly PM procedures, and automation metadata. Source reference: uploaded LINE-2-PACKAGING manual, page 1.

The source explicitly states that LINE-2-PACKAGING telemetry thresholds should be evaluated against active IT production batch contexts and provides specific SKU-100, SKU-500, and SKU-899 operating data. Source reference: uploaded LINE-2-PACKAGING manual, page 1.

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Global Operating Standard - Telemetry and SKU Context

The following cross-line guardrails are the required baseline for this knowledge base and are repeated in each line section to improve retrieval precision.

Parameter	Required cross-line baseline	Warning / action threshold	AI interpretation note
Temperature	Around 75.0 °C	Critical upper warning: 85.0 °C	Persistently above 85.0 °C should trigger immediate review, with MONITOR_EQUIPMENT and feed/load reduction as applicable.
Vibration	Around 2.0 RMS	Critical upper warning: 2.35 RMS	Sustained or repeating values above 2.35 RMS should trigger inspection. Evaluate against batch SKU and changeover context.

Sensor normalization: Temperature is expressed in °C. Vibration is expressed as RMS in the telemetry system; when a line-specific source uses another explicit unit such as mm/s, retain that source unit in the line-specific diagnostic context rather than silently converting it.

SKU-specific behavior

SKU	Nominal contextual behavior	Expected temperature response	Expected vibration response	RUL / interpretation
SKU-100	Standard / high-speed recipe context; use production schedule to confirm batch phase.	Baseline around 75.0 °C unless other load conditions apply.	Baseline around 2.0 RMS.	Unexpected vibration during a stable SKU-100 run is a strong mechanical anomaly indicator, especially when above 2.35 RMS.
SKU-500	Mid-load production context; adhesive or material buildup may increase drive load on relevant equipment.	Use 75.0 °C baseline unless telemetry shows recipe-specific deviation.	Use 2.0 RMS baseline unless telemetry shows recipe-specific deviation.	Watch cumulative stress and maintenance history; repeated buildup events can increase mechanical load.
SKU-899	Heavy-duty / high-stress context. Known to raise mechanical and thermal loading.	15% increase from 75.0 °C = approx. 86.25 °C.	20% increase from 2.0 RMS = approx. 2.4 RMS.	Prolonged processing degrades RUL faster due to mechanical stress. Treat duration, trend, and recurrence as key factors.

Machine Changeover context: Changeovers typically take 30 minutes between batches. Frequent changeovers contribute to mechanical wear and tear and can introduce calibration drift. The AI agent should classify changeover-related sensor excursions separately from steady-state production excursions, then verify whether the elevated signal persists after the line returns to stable production.

RUL interpretation rules

- **RUL is a trend signal, not a calendar date.** A declining RUL estimate should be interpreted with recent vibration/temperature trajectory, load, SKU exposure, operating hours, and maintenance history.
- **SKU-899 exposure accelerates degradation.** Prolonged processing of SKU-899 is known to increase thermal and mechanical loading, so cumulative time at that load should be included in the failure-risk context.
- **Changeover count matters.** Frequent 30-minute Machine Changeovers introduce additional mechanical cycling and can cause calibration drift; repeated post-changeover baseline shifts should increase maintenance priority.

Standard severity logic for RAG agents

Condition	Suggested classification	Required next step
Below warning and stable trend	Normal operating context	Continue monitoring and maintain routine PM.
Near warning, rising trend, or repeated excursions	Watch / investigate	MONITOR_EQUIPMENT; correlate with SKU, speed, load, changeover state, and recent maintenance.
Temperature > 85.0 °C or vibration > 2.35 RMS for sustained/repeating periods	Warning condition	Reduce load/feed when safe; inspect likely mechanical/thermal causes; schedule maintenance as indicated.
Low RUL plus persistent warning telemetry	High maintenance priority	Schedule preventative maintenance during the next changeover where feasible; create work order when integrated.
Critical anomaly relative to active SKU or safety system	Engineering / safety escalation	Follow site escalation and LOTO procedures; do not rely on automated classification alone.

LINE-2-PACKAGING - Source Context and Guardrail Reconciliation

The uploaded LINE-2-PACKAGING source contains equipment-specific thresholds that are more granular than the cross-line guardrails required for this manual. Those source values are retained here as a distinct packaging-equipment context.

LINE-2 source parameter	SKU-100	SKU-500	SKU-899
Nominal speed	450 units/min	320 units/min	180 units/min
Target OEE baseline	88.0%	85.0%	80.0%
Normal vibration range	0.8-1.8 mm/s	1.2-2.2 mm/s	2.0-3.2 mm/s
Max continuous vibration	2.5 mm/s	3.0 mm/s	4.2 mm/s
Normal bearing temperature	45-62 °C	50-68 °C	55-75 °C
Thermal warning cutoff	70 °C	75 °C	82 °C

These values are sourced from page 1 of the uploaded LINE-2-PACKAGING manual. The source also states that SKU-899 vibration up to 3.2 mm/s is baseline-normal for that packaging drive context, but becomes a critical anomaly if observed during SKU-100. Source reference: uploaded LINE-2-PACKAGING manual, page 1.

Interpretation rule: The cross-line 75.0 °C / 85.0 °C and 2.0 RMS / 2.35 RMS guardrails are the mandatory standardized values for this knowledge base. The LINE-2 source values should only be retrieved as additional packaging-machine context when the telemetry tag is mapped to that specific equipment and unit convention. Do not substitute one table for the other.

LINE-1-MIXING - Maintenance and Operations Reference

Line ID: LINE-1-MIXING

Functional scope: The upstream process line that combines approved raw materials to a controlled mixture before downstream filling and packaging. The functional equipment block typically includes a mixing vessel or blender, drive motor, agitator or impeller, inlet/feed devices, cooling provisions, and instrumentation for temperature and vibration monitoring.

Line operating profile

High torque during startup, ingredient addition, high-viscosity portions of a batch, and extended mixing cycles. Mechanical loading may vary significantly by recipe and batch sequence.

Major functional equipment blocks

Mixing vessel/blender; agitator/impeller; main drive motor and coupling; gearbox or reduction stage; feed/inlet equipment; cooling circuit; temperature sensor; vibration sensor; control panel/VFD.

1. Equipment Overview & Purpose

The upstream process line that combines approved raw materials to a controlled mixture before downstream filling and packaging. The functional equipment block typically includes a mixing vessel or blender, drive motor, agitator or impeller, inlet/feed devices, cooling provisions, and instrumentation for temperature and vibration monitoring.

Operational objective: Maintain stable throughput and product quality while keeping temperature, vibration, machine condition, and changeover calibration within expected behavior for the active SKU and batch phase.

2. Normal Operating Ranges

Parameter	Required cross-line baseline	Warning / action threshold	AI interpretation note
Temperature	Around 75.0 °C	Critical upper warning: 85.0 °C	Persistently above 85.0 °C should trigger immediate review, with MONITOR_EQUIPMENT and feed/load reduction as applicable.
Vibration	Around 2.0 RMS	Critical upper warning: 2.35 RMS	Sustained or repeating values above 2.35 RMS should trigger inspection. Evaluate against batch SKU and changeover context.

Context note for SKU-899: The standardized baseline predicts approximately 86.25 °C and 2.4 RMS during SKU-899. These values exceed the cross-line warning thresholds by design, so the AI agent must distinguish expected SKU-899 load context from an uncommanded failure condition. Persistence, rate of rise, duration, and failure to return toward baseline after the SKU or batch phase changes should increase concern.

3. Stress Factors & SKU-Specific Behavior

SKU	Nominal contextual behavior	Expected temperature response	Expected vibration response	RUL / interpretation
SKU-100	Standard / high-speed recipe context; use production schedule to confirm batch phase.	Baseline around 75.0 °C unless other load conditions apply.	Baseline around 2.0 RMS.	Unexpected vibration during a stable SKU-100 run is a strong mechanical anomaly indicator, especially when above 2.35 RMS.
SKU-500	Mid-load production context; adhesive or material buildup may increase drive load on relevant equipment.	Use 75.0 °C baseline unless telemetry shows recipe-specific deviation.	Use 2.0 RMS baseline unless telemetry shows recipe-specific deviation.	Watch cumulative stress and maintenance history; repeated buildup events can increase mechanical load.
SKU-899	Heavy-duty / high-stress context. Known to raise mechanical and thermal loading.	15% increase from 75.0 °C = approx. 86.25 °C.	20% increase from 2.0 RMS = approx. 2.4 RMS.	Prolonged processing degrades RUL faster due to mechanical stress. Treat duration, trend, and recurrence as key factors.

Machine Changeover context: Changeovers typically take 30 minutes between batches. Frequent changeovers contribute to mechanical wear and tear and can introduce calibration drift. The AI agent should classify changeover-related sensor excursions separately from steady-state production excursions, then verify whether the elevated signal persists after the line returns to stable production.

Line-specific stress factors

Viscosity swings, solids loading, off-center material accumulation, impeller imbalance, coupling wear, gearbox lubrication condition, frequent starts/stops, and repeated changeovers.

RUL impact: Prolonged high-load SKU-899 exposure accelerates mechanical wear because the line operates under increased torque, thermal load, or cycle stress. A predictive model should therefore use cumulative SKU-899 run duration and the number of repeated high-stress batches as explanatory features rather than treating each batch independently.

4. Recommended Maintenance Actions

Observed symptom / trigger	Likely interpretation	Recommended action	AI / RAG action cue
High vibration during Machine Changeover	Mechanical alignment drift, coupling movement, loose fasteners, bearing stress, or calibration drift after format/tool change.	Inspect bearings; inspect couplings and fasteners; verify mechanical alignment; re-calibrate motor alignment; capture post-changeover baseline.	MONITOR_EQUIPME NT; compare with pre-changeover and post-changeover baselines.
High vibration during SKU-899 run	Expected load increase may be present, but persistent values above the required 2.35 RMS warning indicate elevated mechanical stress.	Inspect bearings; check drive/gearbox condition; re-calibrate motor alignment; verify load path and fasteners; assess trend against batch duration.	MONITOR_EQUIPME NT; escalate when vibration remains elevated after stable production conditions return.
Temperature exceeds 85.0 °C	Thermal load, cooling deficiency, lubrication problem, excessive feed/load, or sustained high-duty operation.	Check coolant levels / cooling flow where fitted; verify lubrication; reduce feed rate; check airflow or thermal path; verify sensor plausibility.	MONITOR_EQUIPME NT; correlate with SKU, line speed, and batch age.
Low Predicted RUL	Cumulative stress or degradation indicates maintenance should be planned before the next high-load operating window.	Schedule preventative maintenance during next changeover; inspect wear components; document condition and reset baseline only after verified repair.	SCHEDULE_PM / create_work_order when integrated; respect LOTO controls.
Temperature and vibration rise together	Joint thermal-mechanical deterioration is more concerning than either signal alone; possible bearing, alignment, lubrication, or overload issue.	Reduce load if safe; inspect bearings, alignment, lubrication, and cooling; escalate for engineering review if trend continues.	Raise diagnostic priority based on persistence and rate of rise.
High reject rate or micro-stoppages with stable temperature	Functional alignment, timing, pressure, sensor, or actuator issue may precede a thermal/mechanical failure.	Check alignment, actuator pressure, sensors, guides, and timing; review recent changeover record.	Cross-reference maintenance history and batch schedule before classifying as mechanical failure.

Line-specific troubleshooting focus

Symptom	Line-specific checks	Escalation cue
High vibration at steady state	Inspect agitator and shaft bearings for heat, noise, and play. Verify motor/gearbox coupling alignment after significant changeovers or abnormal vibration events.	Persistent > 2.35 RMS or rising trend after load stabilization.
Temperature rise	Verify cooling circuit flow and temperature sensor plausibility when thermal alarms occur.	Sustained > 85.0 °C, rapid rate of rise, or coupled vibration increase.
Post-changeover baseline shift	Verify motor/gearbox coupling alignment after significant changeovers or abnormal vibration events. Check impeller/agitator condition for fouling, imbalance, or buildup.	Signal remains elevated after production returns to stable conditions.
Low predicted RUL	Review maintenance history, cumulative high-load exposure, and current telemetry; plan inspection before next heavy-duty batch where feasible.	RUL low plus warning telemetry or increasing trend.

5. Maintenance Protocols & Inspection Schedule

Interval	Maintenance activity	RAG planning rule
Per shift / daily	Review telemetry trend, inspect visible leaks/loose hardware, verify guards/interlocks, and confirm no unacknowledged alarms.	Capture baseline after maintenance or changeover.
Weekly	Physical inspection of bearings, couplings, guards, sensors, pneumatic/hydraulic or cooling provisions as fitted.	Compare measured values with prior baseline and current batch context.
Monthly	Comprehensive alignment, lubrication, fastener, sensor, and electrical checks appropriate to the line equipment.	Use RUL to prioritize work before predicted degradation windows.
After repeated changeovers	Re-check calibration, alignment, reference positions, and sensor zero/span where applicable.	Flag calibration drift when post-changeover baseline fails to return.

6. Telemetry + IT Batch Schedule Correlation

Prioritize temperature and vibration trends against batch phase. Distinguish startup transients, ingredient addition peaks, steady-state mixing, and post-mix drain/cleaning periods.

Telemetry feature	Schedule / state feature	Recommended interpretation
Temperature trend	Active SKU + batch elapsed time	Look for persistent upward drift versus expected batch behavior.
Vibration trend	Machine Changeover flag + changeover duration	Separate changeover transients from production-state baseline.
Spike frequency	Batch count + recipe sequence	Repeated spikes across batches suggest degradation rather than an isolated event.
Predicted RUL	Upcoming schedule / next changeover	Use maintenance window proximity to time preventative work without disrupting high-risk production.
OEE / reject / micro-stop indicators	SKU and line speed	Cross-check quality/performance symptoms against mechanical telemetry before escalating hardware failure.

7. Standard AI Recommendation Playbooks

Trigger pattern	Recommended response sequence
Temperature > 85.0 °C	1) MONITOR_EQUIPMENT. 2) Correlate with SKU, speed, and load. 3) Check coolant/cooling and lubrication. 4) Reduce feed rate or load when safe. 5) Schedule inspection if the condition persists.
Vibration > 2.35 RMS during changeover or SKU-899	1) Preserve changeover context. 2) Inspect bearings and coupling/drive alignment. 3) Re-calibrate motor alignment. 4) Compare post-changeover baseline. 5) Escalate if elevated vibration persists in steady state.
Low Predicted RUL	1) Compare RUL trend to recent warnings. 2) Identify next changeover window. 3) Schedule preventative maintenance during the next changeover. 4) Create work order and LOTO checklist when integrated.
Repeated warning across multiple batches	Treat as a degradation trend. Increase maintenance priority, inspect the common failure path, and record confirmed findings so future retrieval can distinguish this failure mode from transient SKU effects.

8. Automation & Agent Resolution Metadata

The metadata block below is intended for indexing and deterministic retrieval. Key strings are intentionally repeated exactly, including the line ID, SKUs, and maintenance action token MONITOR_EQUIPMENT.

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{
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    "SKU-500",
    "SKU-899",
    "Machine Changeover",
    "RUL",
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    "InvestigativeAgent",
    "ExecutionAgent"
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    "create_work_order",
    "trigger_loto_checklist"
  ]
}
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LINE-2-PACKAGING - Maintenance and Operations Reference

Line ID: LINE-2-PACKAGING

Functional scope: The automated high-speed packaging line transfers, forms, seals or labels, and discharges finished units at controlled throughput. The uploaded OEM source defines this line as operating under variable mechanical stress governed by active SKU characteristics and requires vibration and bearing temperature to be evaluated against IT batch context.

Line operating profile

High speed for SKU-100, moderate throughput for SKU-500, and high-torque/heavy-duty operation for SKU-899. The source document reports nominal speeds of 450 units/min, 320 units/min, and 180 units/min respectively.

Major functional equipment blocks

Main drive shaft; Motor M-1/M-2; drive bearings including B-12; Poly-V belt; drive rollers; VFD modules; pneumatic regulator and lines; sealing jaw assembly; heating element H-1; optical sensors S-1 through S-8.

1. Equipment Overview & Purpose

The automated high-speed packaging line transfers, forms, seals or labels, and discharges finished units at controlled throughput. The uploaded OEM source defines this line as operating under variable mechanical stress governed by active SKU characteristics and requires vibration and bearing temperature to be evaluated against IT batch context.

Operational objective: Maintain stable throughput and product quality while keeping temperature, vibration, machine condition, and changeover calibration within expected behavior for the active SKU and batch phase.

2. Normal Operating Ranges

Parameter	Required cross-line baseline	Warning / action threshold	AI interpretation note
Temperature	Around 75.0 °C	Critical upper warning: 85.0 °C	Persistently above 85.0 °C should trigger immediate review, with MONITOR_EQUIPMENT and feed/load reduction as applicable.
Vibration	Around 2.0 RMS	Critical upper warning: 2.35 RMS	Sustained or repeating values above 2.35 RMS should trigger inspection. Evaluate against batch SKU and changeover context.

Context note for SKU-899: The standardized baseline predicts approximately 86.25 °C and 2.4 RMS during SKU-899. These values exceed the cross-line warning thresholds by design, so the AI agent must distinguish expected SKU-899 load context from an uncommanded failure condition. Persistence, rate of rise, duration, and failure to return toward baseline after the SKU or batch phase changes should increase concern.

For LINE-2-PACKAGING, the source document provides additional equipment-specific operating values (for example, SKU-899 vibration 2.0-3.2 mm/s and temperature 55-75 °C) plus max continuous vibration and thermal cutoffs. Retain the source unit and equipment mapping when using those detailed diagnostic thresholds. Source reference: uploaded LINE-2-PACKAGING manual, page 1.

3. Stress Factors & SKU-Specific Behavior

SKU	Nominal contextual behavior	Expected temperature response	Expected vibration response	RUL / interpretation
SKU-100	Standard / high-speed recipe context; use production schedule to confirm batch phase.	Baseline around 75.0 °C unless other load conditions apply.	Baseline around 2.0 RMS.	Unexpected vibration during a stable SKU-100 run is a strong mechanical anomaly indicator, especially when above 2.35 RMS.
SKU-500	Mid-load production context; adhesive or material buildup may increase drive load on relevant equipment.	Use 75.0 °C baseline unless telemetry shows recipe-specific deviation.	Use 2.0 RMS baseline unless telemetry shows recipe-specific deviation.	Watch cumulative stress and maintenance history; repeated buildup events can increase mechanical load.
SKU-899	Heavy-duty / high-stress context. Known to raise mechanical and thermal loading.	15% increase from 75.0 °C = approx. 86.25 °C.	20% increase from 2.0 RMS = approx. 2.4 RMS.	Prolonged processing degrades RUL faster due to mechanical stress. Treat duration, trend, and recurrence as key factors.

Machine Changeover context: Changeovers typically take 30 minutes between batches. Frequent changeovers contribute to mechanical wear and tear and can introduce calibration drift. The AI agent should classify changeover-related sensor excursions separately from steady-state production excursions, then verify whether the elevated signal persists after the line returns to stable production.

Line-specific stress factors

Sustained torque on long SKU-500/899 runs, adhesive/resin buildup from SKU-500, high rotational speed and grease shear during SKU-100, shaft misalignment following SKU changeovers, pneumatic pressure fluctuations, and SKU-899 sealing thermal mismatch.

RUL impact: Prolonged high-load SKU-899 exposure accelerates mechanical wear because the line operates under increased torque, thermal load, or cycle stress. A predictive model should therefore use cumulative SKU-899 run duration and the number of repeated high-stress batches as explanatory features rather than treating each batch independently.

4. Recommended Maintenance Actions

Observed symptom / trigger	Likely interpretation	Recommended action	AI / RAG action cue
High vibration during Machine Changeover	Mechanical alignment drift, coupling movement, loose fasteners, bearing stress, or calibration drift after format/tool change.	Inspect bearings; inspect couplings and fasteners; verify mechanical alignment; re-calibrate motor alignment; capture post-changeover baseline.	MONITOR_EQUIPME NT; compare with pre-changeover and post-changeover baselines.
High vibration during SKU-899 run	Expected load increase may be present, but persistent values above the required 2.35 RMS warning indicate elevated mechanical stress.	Inspect bearings; check drive/gearbox condition; re-calibrate motor alignment; verify load path and fasteners; assess trend against batch duration.	MONITOR_EQUIPME NT; escalate when vibration remains elevated after stable production conditions return.
Temperature exceeds 85.0 °C	Thermal load, cooling deficiency, lubrication problem, excessive feed/load, or sustained high-duty operation.	Check coolant levels / cooling flow where fitted; verify lubrication; reduce feed rate; check airflow or thermal path; verify sensor plausibility.	MONITOR_EQUIPME NT; correlate with SKU, line speed, and batch age.
Low Predicted RUL	Cumulative stress or degradation indicates maintenance should be planned before the next high-load operating window.	Schedule preventative maintenance during next changeover; inspect wear components; document condition and reset baseline only after verified repair.	SCHEDULE_PM / create_work_order when integrated; respect LOTO controls.
Temperature and vibration rise together	Joint thermal-mechanical deterioration is more concerning than either signal alone; possible bearing, alignment, lubrication, or overload issue.	Reduce load if safe; inspect bearings, alignment, lubrication, and cooling; escalate for engineering review if trend continues.	Raise diagnostic priority based on persistence and rate of rise.
High reject rate or micro-stoppages with stable temperature	Functional alignment, timing, pressure, sensor, or actuator issue may precede a thermal/mechanical failure.	Check alignment, actuator pressure, sensors, guides, and timing; review recent changeover record.	Cross-reference maintenance history and batch schedule before classifying as mechanical failure.

Line-specific troubleshooting focus

Symptom	Line-specific checks	Escalation cue
High vibration at steady state	Inspect Poly-V belt tension with sonic tension meter; source target is 140 N +/- 5 N. Clean drive rollers using IPA when debris or resin is present.	Persistent > 2.35 RMS or rising trend after load stabilization.
Temperature rise	Check drive sprocket alignment and chain slack; source allowance is 10-15 mm deflection.	Sustained > 85.0 °C, rapid rate of rise, or coupled vibration increase.
Post-changeover baseline shift	Clean drive rollers using IPA when debris or resin is present. Retighten tensioner bolt B-4 to the source torque of 45 Nm after inspection.	Signal remains elevated after production returns to stable conditions.
Low predicted RUL	Review maintenance history, cumulative high-load exposure, and current telemetry; plan inspection before next heavy-duty batch where feasible.	RUL low plus warning telemetry or increasing trend.

5. Maintenance Protocols & Inspection Schedule

Interval	Maintenance activity	Source-specific reference
Every 168 operating hours	Clean optical sensors S-1 through S-8; record baseline vibration across all 4 axis channels on Motor M-1 and M-2; inspect pneumatic lines for micro-leaks or pressure drops.	Source PM-W schedule.
Every 720 operating hours	Check primary drive sprocket alignment and chain slack; lubricate drive shaft bearings; inspect VFD electrical connection torque.	Source PM-M: chain slack 10-15 mm; Mobilith SHC 220 12-15 g; VFD torque 3.5 Nm.

6. Telemetry + IT Batch Schedule Correlation

Correlate OEE, bearing vibration, bearing temperature, speed, pneumatic pressure, and batch SKU. The source states that up to 3.2 mm/s vibration can be baseline-normal for SKU-899 in its specific packaging drive context, but the same condition is identified as a critical anomaly when observed on a SKU-100 run.

Telemetry feature	Schedule / state feature	Recommended interpretation
Temperature trend	Active SKU + batch elapsed time	Look for persistent upward drift versus expected batch behavior.
Vibration trend	Machine Changeover flag + changeover duration	Separate changeover transients from production-state baseline.
Spike frequency	Batch count + recipe sequence	Repeated spikes across batches suggest degradation rather than an isolated event.
Predicted RUL	Upcoming schedule / next changeover	Use maintenance window proximity to time preventative work without disrupting high-risk production.
OEE / reject / micro-stop indicators	SKU and line speed	Cross-check quality/performance symptoms against mechanical telemetry before escalating hardware failure.

7. Standard AI Recommendation Playbooks

Trigger pattern	Recommended response sequence
Temperature > 85.0 °C	1) MONITOR_EQUIPMENT. 2) Correlate with SKU, speed, and load. 3) Check coolant/cooling and lubrication. 4) Reduce feed rate or load when safe. 5) Schedule inspection if the condition persists.
Vibration > 2.35 RMS during changeover or SKU-899	1) Preserve changeover context. 2) Inspect bearings and coupling/drive alignment. 3) Re-calibrate motor alignment. 4) Compare post-changeover baseline. 5) Escalate if elevated vibration persists in steady state.
Low Predicted RUL	1) Compare RUL trend to recent warnings. 2) Identify next changeover window. 3) Schedule preventative maintenance during the next changeover. 4) Create work order and LOTO checklist when integrated.
Repeated warning across multiple batches	Treat as a degradation trend. Increase maintenance priority, inspect the common failure path, and record confirmed findings so future retrieval can distinguish this failure mode from transient SKU effects.

8. Automation & Agent Resolution Metadata

The metadata block below is intended for indexing and deterministic retrieval. Key strings are intentionally repeated exactly, including the line ID, SKUs, and maintenance action token MONITOR_EQUIPMENT.

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}
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Uploaded-source metadata: The original LINE-2-PACKAGING manual also includes target agents DiagnosticAgent, InvestigativeAgent, ExecutionAgent and MCP actions slack_alert_maintenance, create_work_order, trigger_loto_checklist. These are preserved as part of the source context. Source reference: uploaded LINE-2-PACKAGING manual, page 2.

LINE-3-FILLING - Maintenance and Operations Reference

Line ID: LINE-3-FILLING

Functional scope: The filling line meters and transfers product into primary containers at the required fill volume and production rate. The functional equipment block typically includes dosing pumps or augers, product manifold/valves, filling heads/nozzles, conveyors, actuators, and temperature/vibration instrumentation.

Line operating profile

Cyclic loading from repeated fill strokes, pump starts/stops, valve actuation, and conveyor motion. Product viscosity and target fill volume influence motor load and cycle stability.

Major functional equipment blocks

Filling pump or dosing mechanism; drive motor; coupling/gearbox; product valves; manifold; filling heads/nozzles; conveyor; actuators; temperature sensor; vibration sensor; HMI/PLC/VFD.

1. Equipment Overview & Purpose

The filling line meters and transfers product into primary containers at the required fill volume and production rate. The functional equipment block typically includes dosing pumps or augers, product manifold/valves, filling heads/nozzles, conveyors, actuators, and temperature/vibration instrumentation.

Operational objective: Maintain stable throughput and product quality while keeping temperature, vibration, machine condition, and changeover calibration within expected behavior for the active SKU and batch phase.

2. Normal Operating Ranges

Parameter	Required cross-line baseline	Warning / action threshold	AI interpretation note
Temperature	Around 75.0 °C	Critical upper warning: 85.0 °C	Persistently above 85.0 °C should trigger immediate review, with MONITOR_EQUIPMENT and feed/load reduction as applicable.
Vibration	Around 2.0 RMS	Critical upper warning: 2.35 RMS	Sustained or repeating values above 2.35 RMS should trigger inspection. Evaluate against batch SKU and changeover context.

Context note for SKU-899: The standardized baseline predicts approximately 86.25 °C and 2.4 RMS during SKU-899. These values exceed the cross-line warning thresholds by design, so the AI agent must distinguish expected SKU-899 load context from an uncommanded failure condition. Persistence, rate of rise, duration, and failure to return toward baseline after the SKU or batch phase changes should increase concern.

3. Stress Factors & SKU-Specific Behavior

SKU	Nominal contextual behavior	Expected temperature response	Expected vibration response	RUL / interpretation
SKU-100	Standard / high-speed recipe context; use production schedule to confirm batch phase.	Baseline around 75.0 °C unless other load conditions apply.	Baseline around 2.0 RMS.	Unexpected vibration during a stable SKU-100 run is a strong mechanical anomaly indicator, especially when above 2.35 RMS.
SKU-500	Mid-load production context; adhesive or material buildup may increase drive load on relevant equipment.	Use 75.0 °C baseline unless telemetry shows recipe-specific deviation.	Use 2.0 RMS baseline unless telemetry shows recipe-specific deviation.	Watch cumulative stress and maintenance history; repeated buildup events can increase mechanical load.
SKU-899	Heavy-duty / high-stress context. Known to raise mechanical and thermal loading.	15% increase from 75.0 °C = approx. 86.25 °C.	20% increase from 2.0 RMS = approx. 2.4 RMS.	Prolonged processing degrades RUL faster due to mechanical stress. Treat duration, trend, and recurrence as key factors.

Machine Changeover context: Changeovers typically take 30 minutes between batches. Frequent changeovers contribute to mechanical wear and tear and can introduce calibration drift. The AI agent should classify changeover-related sensor excursions separately from steady-state production excursions, then verify whether the elevated signal persists after the line returns to stable production.

Line-specific stress factors

Product viscosity, cavitation or restricted product flow, repeated valve cycling, nozzle fouling, pump seal wear, misalignment, conveyor buildup, and repeated recipe or format changeovers.

RUL impact: Prolonged high-load SKU-899 exposure accelerates mechanical wear because the line operates under increased torque, thermal load, or cycle stress. A predictive model should therefore use cumulative SKU-899 run duration and the number of repeated high-stress batches as explanatory features rather than treating each batch independently.

4. Recommended Maintenance Actions

Observed symptom / trigger	Likely interpretation	Recommended action	AI / RAG action cue
High vibration during Machine Changeover	Mechanical alignment drift, coupling movement, loose fasteners, bearing stress, or calibration drift after format/tool change.	Inspect bearings; inspect couplings and fasteners; verify mechanical alignment; re-calibrate motor alignment; capture post-changeover baseline.	MONITOR_EQUIPME NT; compare with pre-changeover and post-changeover baselines.
High vibration during SKU-899 run	Expected load increase may be present, but persistent values above the required 2.35 RMS warning indicate elevated mechanical stress.	Inspect bearings; check drive/gearbox condition; re-calibrate motor alignment; verify load path and fasteners; assess trend against batch duration.	MONITOR_EQUIPME NT; escalate when vibration remains elevated after stable production conditions return.
Temperature exceeds 85.0 °C	Thermal load, cooling deficiency, lubrication problem, excessive feed/load, or sustained high-duty operation.	Check coolant levels / cooling flow where fitted; verify lubrication; reduce feed rate; check airflow or thermal path; verify sensor plausibility.	MONITOR_EQUIPME NT; correlate with SKU, line speed, and batch age.
Low Predicted RUL	Cumulative stress or degradation indicates maintenance should be planned before the next high-load operating window.	Schedule preventative maintenance during next changeover; inspect wear components; document condition and reset baseline only after verified repair.	SCHEDULE_PM / create_work_order when integrated; respect LOTO controls.
Temperature and vibration rise together	Joint thermal-mechanical deterioration is more concerning than either signal alone; possible bearing, alignment, lubrication, or overload issue.	Reduce load if safe; inspect bearings, alignment, lubrication, and cooling; escalate for engineering review if trend continues.	Raise diagnostic priority based on persistence and rate of rise.
High reject rate or micro-stoppages with stable temperature	Functional alignment, timing, pressure, sensor, or actuator issue may precede a thermal/mechanical failure.	Check alignment, actuator pressure, sensors, guides, and timing; review recent changeover record.	Cross-reference maintenance history and batch schedule before classifying as mechanical failure.

Line-specific troubleshooting focus

Symptom	Line-specific checks	Escalation cue
High vibration at steady state	Inspect pump and drive bearings when vibration rises during filling strokes or changeover. Check suction/product path restrictions before replacing mechanical components.	Persistent > 2.35 RMS or rising trend after load stabilization.
Temperature rise	Confirm temperature sensor plausibility and product-side cooling or thermal control where fitted.	Sustained > 85.0 °C, rapid rate of rise, or coupled vibration increase.
Post-changeover baseline shift	Check suction/product path restrictions before replacing mechanical components. Verify valve timing and fill-head alignment after format changeover.	Signal remains elevated after production returns to stable conditions.
Low predicted RUL	Review maintenance history, cumulative high-load exposure, and current telemetry; plan inspection before next heavy-duty batch where feasible.	RUL low plus warning telemetry or increasing trend.

5. Maintenance Protocols & Inspection Schedule

Interval	Maintenance activity	RAG planning rule
Per shift / daily	Review telemetry trend, inspect visible leaks/loose hardware, verify guards/interlocks, and confirm no unacknowledged alarms.	Capture baseline after maintenance or changeover.
Weekly	Physical inspection of bearings, couplings, guards, sensors, pneumatic/hydraulic or cooling provisions as fitted.	Compare measured values with prior baseline and current batch context.
Monthly	Comprehensive alignment, lubrication, fastener, sensor, and electrical checks appropriate to the line equipment.	Use RUL to prioritize work before predicted degradation windows.
After repeated changeovers	Re-check calibration, alignment, reference positions, and sensor zero/span where applicable.	Flag calibration drift when post-changeover baseline fails to return.

6. Telemetry + IT Batch Schedule Correlation

Separate short cyclic peaks from a rising baseline. Compare vibration to cycle position and batch SKU, and compare temperature to motor load, pump duty cycle, and elapsed batch time.

Telemetry feature	Schedule / state feature	Recommended interpretation
Temperature trend	Active SKU + batch elapsed time	Look for persistent upward drift versus expected batch behavior.
Vibration trend	Machine Changeover flag + changeover duration	Separate changeover transients from production-state baseline.
Spike frequency	Batch count + recipe sequence	Repeated spikes across batches suggest degradation rather than an isolated event.
Predicted RUL	Upcoming schedule / next changeover	Use maintenance window proximity to time preventative work without disrupting high-risk production.
OEE / reject / micro-stop indicators	SKU and line speed	Cross-check quality/performance symptoms against mechanical telemetry before escalating hardware failure.

7. Standard AI Recommendation Playbooks

Trigger pattern	Recommended response sequence
Temperature > 85.0 °C	1) MONITOR_EQUIPMENT. 2) Correlate with SKU, speed, and load. 3) Check coolant/cooling and lubrication. 4) Reduce feed rate or load when safe. 5) Schedule inspection if the condition persists.
Vibration > 2.35 RMS during changeover or SKU-899	1) Preserve changeover context. 2) Inspect bearings and coupling/drive alignment. 3) Re-calibrate motor alignment. 4) Compare post-changeover baseline. 5) Escalate if elevated vibration persists in steady state.
Low Predicted RUL	1) Compare RUL trend to recent warnings. 2) Identify next changeover window. 3) Schedule preventative maintenance during the next changeover. 4) Create work order and LOTO checklist when integrated.
Repeated warning across multiple batches	Treat as a degradation trend. Increase maintenance priority, inspect the common failure path, and record confirmed findings so future retrieval can distinguish this failure mode from transient SKU effects.

8. Automation & Agent Resolution Metadata

The metadata block below is intended for indexing and deterministic retrieval. Key strings are intentionally repeated exactly, including the line ID, SKUs, and maintenance action token MONITOR_EQUIPMENT.

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{
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    "trigger_loto_checklist"
  ]
}
```

LINE-4-SEALING - Maintenance and Operations Reference

Line ID: LINE-4-SEALING

Functional scope: The sealing line closes and secures product containers or packages using heated, pressure-controlled, and synchronized sealing mechanisms. The functional equipment block typically includes sealing jaws, heaters, temperature controllers, pneumatic actuators, pressure regulation, and line drives.

Line operating profile

Repeated high-cycle motion with coupled thermal and pneumatic loads. Seal quality depends on stable jaw alignment, heater control, actuator pressure, and line speed.

Major functional equipment blocks

Seal jaw assembly; heater blocks/elements; PID temperature controller; pneumatic regulator and actuators; drive motor; conveyor; temperature sensor; vibration sensor; pressure transmitter; reject mechanism.

1. Equipment Overview & Purpose

The sealing line closes and secures product containers or packages using heated, pressure-controlled, and synchronized sealing mechanisms. The functional equipment block typically includes sealing jaws, heaters, temperature controllers, pneumatic actuators, pressure regulation, and line drives.

Operational objective: Maintain stable throughput and product quality while keeping temperature, vibration, machine condition, and changeover calibration within expected behavior for the active SKU and batch phase.

2. Normal Operating Ranges

Parameter	Required cross-line baseline	Warning / action threshold	AI interpretation note
Temperature	Around 75.0 °C	Critical upper warning: 85.0 °C	Persistently above 85.0 °C should trigger immediate review, with MONITOR_EQUIPMENT and feed/load reduction as applicable.
Vibration	Around 2.0 RMS	Critical upper warning: 2.35 RMS	Sustained or repeating values above 2.35 RMS should trigger inspection. Evaluate against batch SKU and changeover context.

Context note for SKU-899: The standardized baseline predicts approximately 86.25 °C and 2.4 RMS during SKU-899. These values exceed the cross-line warning thresholds by design, so the AI agent must distinguish expected SKU-899 load context from an uncommanded failure condition. Persistence, rate of rise, duration, and failure to return toward baseline after the SKU or batch phase changes should increase concern.

3. Stress Factors & SKU-Specific Behavior

SKU	Nominal contextual behavior	Expected temperature response	Expected vibration response	RUL / interpretation
SKU-100	Standard / high-speed recipe context; use production schedule to confirm batch phase.	Baseline around 75.0 °C unless other load conditions apply.	Baseline around 2.0 RMS.	Unexpected vibration during a stable SKU-100 run is a strong mechanical anomaly indicator, especially when above 2.35 RMS.
SKU-500	Mid-load production context; adhesive or material buildup may increase drive load on relevant equipment.	Use 75.0 °C baseline unless telemetry shows recipe-specific deviation.	Use 2.0 RMS baseline unless telemetry shows recipe-specific deviation.	Watch cumulative stress and maintenance history; repeated buildup events can increase mechanical load.
SKU-899	Heavy-duty / high-stress context. Known to raise mechanical and thermal loading.	15% increase from 75.0 °C = approx. 86.25 °C.	20% increase from 2.0 RMS = approx. 2.4 RMS.	Prolonged processing degrades RUL faster due to mechanical stress. Treat duration, trend, and recurrence as key factors.

Machine Changeover context: Changeovers typically take 30 minutes between batches. Frequent changeovers contribute to mechanical wear and tear and can introduce calibration drift. The AI agent should classify changeover-related sensor excursions separately from steady-state production excursions, then verify whether the elevated signal persists after the line returns to stable production.

Line-specific stress factors

Thermal cycling, jaw alignment drift, heater degradation, pneumatic pressure variation, coating wear, repeated changeovers, and high-duty SKU-899 heat cycles.

RUL impact: Prolonged high-load SKU-899 exposure accelerates mechanical wear because the line operates under increased torque, thermal load, or cycle stress. A predictive model should therefore use cumulative SKU-899 run duration and the number of repeated high-stress batches as explanatory features rather than treating each batch independently.

4. Recommended Maintenance Actions

Observed symptom / trigger	Likely interpretation	Recommended action	AI / RAG action cue
High vibration during Machine Changeover	Mechanical alignment drift, coupling movement, loose fasteners, bearing stress, or calibration drift after format/tool change.	Inspect bearings; inspect couplings and fasteners; verify mechanical alignment; re-calibrate motor alignment; capture post-changeover baseline.	MONITOR_EQUIPME NT; compare with pre-changeover and post-changeover baselines.
High vibration during SKU-899 run	Expected load increase may be present, but persistent values above the required 2.35 RMS warning indicate elevated mechanical stress.	Inspect bearings; check drive/gearbox condition; re-calibrate motor alignment; verify load path and fasteners; assess trend against batch duration.	MONITOR_EQUIPME NT; escalate when vibration remains elevated after stable production conditions return.
Temperature exceeds 85.0 °C	Thermal load, cooling deficiency, lubrication problem, excessive feed/load, or sustained high-duty operation.	Check coolant levels / cooling flow where fitted; verify lubrication; reduce feed rate; check airflow or thermal path; verify sensor plausibility.	MONITOR_EQUIPME NT; correlate with SKU, line speed, and batch age.
Low Predicted RUL	Cumulative stress or degradation indicates maintenance should be planned before the next high-load operating window.	Schedule preventative maintenance during next changeover; inspect wear components; document condition and reset baseline only after verified repair.	SCHEDULE_PM / create_work_order when integrated; respect LOTO controls.
Temperature and vibration rise together	Joint thermal-mechanical deterioration is more concerning than either signal alone; possible bearing, alignment, lubrication, or overload issue.	Reduce load if safe; inspect bearings, alignment, lubrication, and cooling; escalate for engineering review if trend continues.	Raise diagnostic priority based on persistence and rate of rise.
High reject rate or micro-stoppages with stable temperature	Functional alignment, timing, pressure, sensor, or actuator issue may precede a thermal/mechanical failure.	Check alignment, actuator pressure, sensors, guides, and timing; review recent changeover record.	Cross-reference maintenance history and batch schedule before classifying as mechanical failure.

Line-specific troubleshooting focus

Symptom	Line-specific checks	Escalation cue
High vibration at steady state	Verify pneumatic regulator stability and actuator response when seal quality changes. Re-check jaw parallelism and mechanical alignment after repeated changeovers.	Persistent > 2.35 RMS or rising trend after load stabilization.
Temperature rise	Inspect seal jaw coating/wear strips and replace when site-defined wear criteria are exceeded.	Sustained > 85.0 °C, rapid rate of rise, or coupled vibration increase.
Post-changeover baseline shift	Re-check jaw parallelism and mechanical alignment after repeated changeovers. Inspect heater and temperature-controller response for overshoot or sluggish recovery.	Signal remains elevated after production returns to stable conditions.
Low predicted RUL	Review maintenance history, cumulative high-load exposure, and current telemetry; plan inspection before next heavy-duty batch where feasible.	RUL low plus warning telemetry or increasing trend.

5. Maintenance Protocols & Inspection Schedule

Interval	Maintenance activity	RAG planning rule
Per shift / daily	Review telemetry trend, inspect visible leaks/loose hardware, verify guards/interlocks, and confirm no unacknowledged alarms.	Capture baseline after maintenance or changeover.
Weekly	Physical inspection of bearings, couplings, guards, sensors, pneumatic/hydraulic or cooling provisions as fitted.	Compare measured values with prior baseline and current batch context.
Monthly	Comprehensive alignment, lubrication, fastener, sensor, and electrical checks appropriate to the line equipment.	Use RUL to prioritize work before predicted degradation windows.
After repeated changeovers	Re-check calibration, alignment, reference positions, and sensor zero/span where applicable.	Flag calibration drift when post-changeover baseline fails to return.

6. Telemetry + IT Batch Schedule Correlation

Use seal reject rate, pressure, heater temperature, line speed, vibration, and batch context together. A thermal alarm with stable vibration may indicate a control or cooling issue rather than a mechanical bearing issue.

Telemetry feature	Schedule / state feature	Recommended interpretation
Temperature trend	Active SKU + batch elapsed time	Look for persistent upward drift versus expected batch behavior.
Vibration trend	Machine Changeover flag + changeover duration	Separate changeover transients from production-state baseline.
Spike frequency	Batch count + recipe sequence	Repeated spikes across batches suggest degradation rather than an isolated event.
Predicted RUL	Upcoming schedule / next changeover	Use maintenance window proximity to time preventative work without disrupting high-risk production.
OEE / reject / micro-stop indicators	SKU and line speed	Cross-check quality/performance symptoms against mechanical telemetry before escalating hardware failure.

7. Standard AI Recommendation Playbooks

Trigger pattern	Recommended response sequence
Temperature > 85.0 °C	1) MONITOR_EQUIPMENT. 2) Correlate with SKU, speed, and load. 3) Check coolant/cooling and lubrication. 4) Reduce feed rate or load when safe. 5) Schedule inspection if the condition persists.
Vibration > 2.35 RMS during changeover or SKU-899	1) Preserve changeover context. 2) Inspect bearings and coupling/drive alignment. 3) Re-calibrate motor alignment. 4) Compare post-changeover baseline. 5) Escalate if elevated vibration persists in steady state.
Low Predicted RUL	1) Compare RUL trend to recent warnings. 2) Identify next changeover window. 3) Schedule preventative maintenance during the next changeover. 4) Create work order and LOTO checklist when integrated.
Repeated warning across multiple batches	Treat as a degradation trend. Increase maintenance priority, inspect the common failure path, and record confirmed findings so future retrieval can distinguish this failure mode from transient SKU effects.

8. Automation & Agent Resolution Metadata

The metadata block below is intended for indexing and deterministic retrieval. Key strings are intentionally repeated exactly, including the line ID, SKUs, and maintenance action token MONITOR_EQUIPMENT.

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{
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    "temperature",
    "SKU-100",
    "SKU-500",
    "SKU-899",
    "Machine Changeover",
    "RUL",
    "MONITOR_EQUIPMENT"
  ],
  "target_agents": [
    "DiagnosticAgent",
    "InvestigativeAgent",
    "ExecutionAgent"
  ],
  "mcp_actions": [
    "slack_alert_maintenance",
    "create_work_order",
    "trigger_loto_checklist"
  ]
}
```

LINE-5-PALLETIZING - Maintenance and Operations Reference

Line ID: LINE-5-PALLETIZING

Functional scope: The end-of-line palletizing process groups finished cases or units and places them into defined pallet patterns for storage or shipment. The functional equipment block may include a robotic or Cartesian palletizer, infeed/outfeed conveyors, grippers or layer handling, vertical/lateral axes, and drive gearboxes.

Line operating profile

High inertial loads during acceleration/deceleration, repeated pick-and-place cycles, and rapid direction changes. Load varies with case weight, pallet pattern, layer count, and cycle rate.

Major functional equipment blocks

Palletizer robot/gantry; servo motors and drives; gearboxes/reducers; end effector/gripper; conveyors; pallet dispenser; safety interlocks; temperature sensor; vibration sensor; control system.

1. Equipment Overview & Purpose

The end-of-line palletizing process groups finished cases or units and places them into defined pallet patterns for storage or shipment. The functional equipment block may include a robotic or Cartesian palletizer, infeed/outfeed conveyors, grippers or layer handling, vertical/lateral axes, and drive gearboxes.

Operational objective: Maintain stable throughput and product quality while keeping temperature, vibration, machine condition, and changeover calibration within expected behavior for the active SKU and batch phase.

2. Normal Operating Ranges

Parameter	Required cross-line baseline	Warning / action threshold	AI interpretation note
Temperature	Around 75.0 °C	Critical upper warning: 85.0 °C	Persistently above 85.0 °C should trigger immediate review, with MONITOR_EQUIPMENT and feed/load reduction as applicable.
Vibration	Around 2.0 RMS	Critical upper warning: 2.35 RMS	Sustained or repeating values above 2.35 RMS should trigger inspection. Evaluate against batch SKU and changeover context.

Context note for SKU-899: The standardized baseline predicts approximately 86.25 °C and 2.4 RMS during SKU-899. These values exceed the cross-line warning thresholds by design, so the AI agent must distinguish expected SKU-899 load context from an uncommanded failure condition. Persistence, rate of rise, duration, and failure to return toward baseline after the SKU or batch phase changes should increase concern.

3. Stress Factors & SKU-Specific Behavior

SKU	Nominal contextual behavior	Expected temperature response	Expected vibration response	RUL / interpretation
SKU-100	Standard / high-speed recipe context; use production schedule to confirm batch phase.	Baseline around 75.0 °C unless other load conditions apply.	Baseline around 2.0 RMS.	Unexpected vibration during a stable SKU-100 run is a strong mechanical anomaly indicator, especially when above 2.35 RMS.
SKU-500	Mid-load production context; adhesive or material buildup may increase drive load on relevant equipment.	Use 75.0 °C baseline unless telemetry shows recipe-specific deviation.	Use 2.0 RMS baseline unless telemetry shows recipe-specific deviation.	Watch cumulative stress and maintenance history; repeated buildup events can increase mechanical load.
SKU-899	Heavy-duty / high-stress context. Known to raise mechanical and thermal loading.	15% increase from 75.0 °C = approx. 86.25 °C.	20% increase from 2.0 RMS = approx. 2.4 RMS.	Prolonged processing degrades RUL faster due to mechanical stress. Treat duration, trend, and recurrence as key factors.

Machine Changeover context: Changeovers typically take 30 minutes between batches. Frequent changeovers contribute to mechanical wear and tear and can introduce calibration drift. The AI agent should classify changeover-related sensor excursions separately from steady-state production excursions, then verify whether the elevated signal persists after the line returns to stable production.

Line-specific stress factors

High acceleration, gear reducer wear, loose fasteners, conveyor chain wear, end-effector imbalance, pallet pattern changes, and repeated changeover calibration events.

RUL impact: Prolonged high-load SKU-899 exposure accelerates mechanical wear because the line operates under increased torque, thermal load, or cycle stress. A predictive model should therefore use cumulative SKU-899 run duration and the number of repeated high-stress batches as explanatory features rather than treating each batch independently.

4. Recommended Maintenance Actions

Observed symptom / trigger	Likely interpretation	Recommended action	AI / RAG action cue
High vibration during Machine Changeover	Mechanical alignment drift, coupling movement, loose fasteners, bearing stress, or calibration drift after format/tool change.	Inspect bearings; inspect couplings and fasteners; verify mechanical alignment; re-calibrate motor alignment; capture post-changeover baseline.	MONITOR_EQUIPME NT; compare with pre-changeover and post-changeover baselines.
High vibration during SKU-899 run	Expected load increase may be present, but persistent values above the required 2.35 RMS warning indicate elevated mechanical stress.	Inspect bearings; check drive/gearbox condition; re-calibrate motor alignment; verify load path and fasteners; assess trend against batch duration.	MONITOR_EQUIPME NT; escalate when vibration remains elevated after stable production conditions return.
Temperature exceeds 85.0 °C	Thermal load, cooling deficiency, lubrication problem, excessive feed/load, or sustained high-duty operation.	Check coolant levels / cooling flow where fitted; verify lubrication; reduce feed rate; check airflow or thermal path; verify sensor plausibility.	MONITOR_EQUIPME NT; correlate with SKU, line speed, and batch age.
Low Predicted RUL	Cumulative stress or degradation indicates maintenance should be planned before the next high-load operating window.	Schedule preventative maintenance during next changeover; inspect wear components; document condition and reset baseline only after verified repair.	SCHEDULE_PM / create_work_order when integrated; respect LOTO controls.
Temperature and vibration rise together	Joint thermal-mechanical deterioration is more concerning than either signal alone; possible bearing, alignment, lubrication, or overload issue.	Reduce load if safe; inspect bearings, alignment, lubrication, and cooling; escalate for engineering review if trend continues.	Raise diagnostic priority based on persistence and rate of rise.
High reject rate or micro-stoppages with stable temperature	Functional alignment, timing, pressure, sensor, or actuator issue may precede a thermal/mechanical failure.	Check alignment, actuator pressure, sensors, guides, and timing; review recent changeover record.	Cross-reference maintenance history and batch schedule before classifying as mechanical failure.

Line-specific troubleshooting focus

Symptom	Line-specific checks	Escalation cue
High vibration at steady state	Inspect reducer, motor, and conveyor bearings when vibration rises during acceleration or deceleration. Verify servo/motor alignment and mechanical fastener torque after changeovers.	Persistent > 2.35 RMS or rising trend after load stabilization.
Temperature rise	Confirm gearbox lubrication and cooling condition when temperature trends upward.	Sustained > 85.0 °C, rapid rate of rise, or coupled vibration increase.
Post-changeover baseline shift	Verify servo/motor alignment and mechanical fastener torque after changeovers. Check end-effector/gripper balance and mounting.	Signal remains elevated after production returns to stable conditions.
Low predicted RUL	Review maintenance history, cumulative high-load exposure, and current telemetry; plan inspection before next heavy-duty batch where feasible.	RUL low plus warning telemetry or increasing trend.

5. Maintenance Protocols & Inspection Schedule

Interval	Maintenance activity	RAG planning rule
Per shift / daily	Review telemetry trend, inspect visible leaks/loose hardware, verify guards/interlocks, and confirm no unacknowledged alarms.	Capture baseline after maintenance or changeover.
Weekly	Physical inspection of bearings, couplings, guards, sensors, pneumatic/hydraulic or cooling provisions as fitted.	Compare measured values with prior baseline and current batch context.
Monthly	Comprehensive alignment, lubrication, fastener, sensor, and electrical checks appropriate to the line equipment.	Use RUL to prioritize work before predicted degradation windows.
After repeated changeovers	Re-check calibration, alignment, reference positions, and sensor zero/span where applicable.	Flag calibration drift when post-changeover baseline fails to return.

6. Telemetry + IT Batch Schedule Correlation

Correlate vibration spikes to axis position, acceleration/deceleration phases, pick/place commands, and batch schedule. Do not treat single-cycle peaks as equivalent to a persistent baseline shift.

Telemetry feature	Schedule / state feature	Recommended interpretation
Temperature trend	Active SKU + batch elapsed time	Look for persistent upward drift versus expected batch behavior.
Vibration trend	Machine Changeover flag + changeover duration	Separate changeover transients from production-state baseline.
Spike frequency	Batch count + recipe sequence	Repeated spikes across batches suggest degradation rather than an isolated event.
Predicted RUL	Upcoming schedule / next changeover	Use maintenance window proximity to time preventative work without disrupting high-risk production.
OEE / reject / micro-stop indicators	SKU and line speed	Cross-check quality/performance symptoms against mechanical telemetry before escalating hardware failure.

7. Standard AI Recommendation Playbooks

Trigger pattern	Recommended response sequence
Temperature > 85.0 °C	1) MONITOR_EQUIPMENT. 2) Correlate with SKU, speed, and load. 3) Check coolant/cooling and lubrication. 4) Reduce feed rate or load when safe. 5) Schedule inspection if the condition persists.
Vibration > 2.35 RMS during changeover or SKU-899	1) Preserve changeover context. 2) Inspect bearings and coupling/drive alignment. 3) Re-calibrate motor alignment. 4) Compare post-changeover baseline. 5) Escalate if elevated vibration persists in steady state.
Low Predicted RUL	1) Compare RUL trend to recent warnings. 2) Identify next changeover window. 3) Schedule preventative maintenance during the next changeover. 4) Create work order and LOTO checklist when integrated.
Repeated warning across multiple batches	Treat as a degradation trend. Increase maintenance priority, inspect the common failure path, and record confirmed findings so future retrieval can distinguish this failure mode from transient SKU effects.

8. Automation & Agent Resolution Metadata

The metadata block below is intended for indexing and deterministic retrieval. Key strings are intentionally repeated exactly, including the line ID, SKUs, and maintenance action token MONITOR_EQUIPMENT.

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    "ExecutionAgent"
  ],
  "mcp_actions": [
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    "create_work_order",
    "trigger_loto_checklist"
  ]
}
```

Appendix A - Cross-Line Maintenance Decision Matrix

Signal / context	Normal / expected context	Warning condition	Maintenance implication
Temperature	Around 75.0 °C; line-specific operating profiles may differ when explicitly mapped.	> 85.0 °C or persistent upward trend.	Check cooling, lubrication, load/feed, sensor plausibility, and process restrictions.
Vibration	Around 2.0 RMS; line-specific machine values may differ when explicitly mapped.	> 2.35 RMS sustained/repeating, or abnormal for active SKU.	Inspect bearings, couplings, alignment, fasteners, gearbox/drive elements, and calibration.
SKU-899	Expected high-stress state with approx. +15% temperature and +20% vibration.	Persistence, worsening trend, or failure to recover after heavy-duty batch.	Account for accelerated RUL degradation and schedule PM around a changeover.
Machine Changeover	Typical duration about 30 minutes.	High vibration/thermal deviation that remains after restart or repeated drift over multiple changeovers.	Inspect alignment and recalibrate motor/reference positions; verify tooling and sensor calibration.
Predicted RUL	Stable or slowly changing under expected load.	Low RUL or rapidly declining RUL concurrent with warnings.	Schedule preventative maintenance during next changeover and create work order when integrated.

Appendix B - RAG Retrieval Patterns

Recommended retrieval keys and semantic patterns include: exact line IDs; active SKU; Machine Changeover; temperature > 85.0 °C; vibration > 2.35 RMS; low Predicted RUL; inspect bearings; re-calibrate motor alignment; check coolant levels; reduce feed rate; MONITOR_EQUIPMENT; schedule preventative maintenance during next changeover; OEE; micro-stoppage; bearing overheating; shaft misalignment; alignment drift; lubrication; cooling; pressure instability.

Retrieval priority: retrieve the exact line section first, then the active SKU behavior, then the symptom-specific troubleshooting row, then maintenance schedule and RUL guidance. When a source-specific threshold exists, preserve its unit and equipment mapping in the generated answer.

Appendix C - LINE-2-PACKAGING Diagnostic Reference Extract

The uploaded packaging source identifies error E-104 as Main Drive Shaft Micro-Stoppage & Speed Degradation, with performance loss and periodic vibration spikes; E-104 root causes include Poly-V belt slippage during long SKU-500/899 runs and drive roller debris from SKU-500 adhesive backing. Corrective actions include LOTO, belt-tension inspection at 140 N +/- 5 N, IPA roller cleaning, and 45 Nm tensioner torque. Source reference: uploaded LINE-2-PACKAGING manual, page 1.

The source identifies E-208 as Bearing Overheating & Thermal Runaway, associated with bearing temperature above 75 °C during SKU-100 and high-frequency vibration above 3.5 mm/s; source root causes include grease degradation and shaft misalignment after SKU changovers. Source reference: uploaded LINE-2-PACKAGING manual, page 1.

The source identifies E-312 as Seal Jaw Misalignment & Micro-Downtime, with high reject rate and micro-stoppages; source causes include pneumatic pressure below 5.8 bar and thermal mismatch during SKU-899 sealing cycles. Corrective actions include verifying 6.2 bar regulator output, re-calibrating PID controller H-1 to 185 °C for SKU-899, and inspecting seal jaw Teflon wear. Source reference: uploaded LINE-2-PACKAGING manual, page 2.

Appendix D - Revision and Data Governance

This edition is a consolidated knowledge-base manual. Any future equipment-specific source should be appended as a controlled context block with its own document ID, line ID, sensor unit conventions, thresholds, error codes, and revision. When source documents conflict, retrieval logic should present the context that matches the exact asset/telemetry tag and document revision rather than merge thresholds without qualification.

End of manual.